

Project Plan: Understanding the Contribution of B-Rep Graph Topology to UV-Net's Strong Performance

Motivation and objective

In the original UV-Net ablation on MFCAD [1], face geometry alone achieved 85.34% IoU, while the full geometry-and-topology model reached 99.87%. Rather than treating this improvement as a single jump, we will rebuild a controlled model ladder to trace the contributions of shallow topology information, learned message passing, joint optimization and edge conditioning. The aim is not a definitive mechanistic explanation, but evidence about the classes and data characteristics for which geometry alone is insufficient and graph context becomes informative.

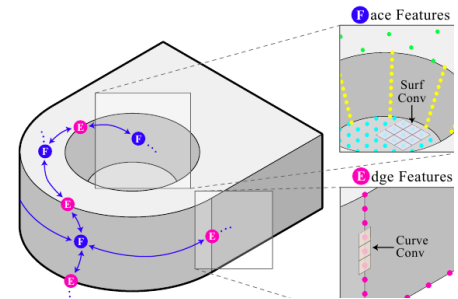


Figure 1. UV-Net architecture combining sampled face and edge geometry through B-Rep graph message passing.

Scope

We focus on MFCAD [2] and its face-level machining-feature classification task. Holding the dataset and task fixed makes changes across stages directly comparable and leaves time for implementation, debugging, exploratory analysis and class-level interpretation.

Controlled model ladder

- 1. Geometry only — face CNN with an independent face classifier.
- 1.5 Geometry + shallow topology — Stage 1 embeddings augmented with explicit Laplacian, rooted Weisfeiler–Lehman and/or graphlet features, without learned message passing.
- 2a. Frozen geometry + GCN — freeze the Stage 1 CNN and train a GCN over its face embeddings.
- 2b. Joint geometry + GCN — train the face CNN and GCN end-to-end.
- 3. Full UV-Net — add edge geometry and edge-conditioned message passing.
- 4. Optional extensions — advanced models such as Graph Transformers.

Analysis and expected contribution

Aggregate performance will quantify each step in the ladder, while per-class metrics and confusion patterns will locate where each gain arises. We will compare learned embeddings across the ladder, visualize selected stages where meaningful changes in class structure or errors emerge, and relate these changes to graph and geometric properties. EDA will examine class balance, graph size and connectivity, surface and edge characteristics, and representative feature structures. This analysis will distinguish contributions from shallow topology, learned message passing, joint optimization and edge conditioning, while identifying cases where geometry alone is sufficient. The intended contribution is an evidence-based decomposition of UV-Net's strong performance on MFCAD.

Project milestones

0. Data readiness — MFCAD loaded, validated and prepared for EDA and model training.
1. Geometry baseline — establish the train/test and evaluation setup; complete the classifier and face CNN (Stage 1), alongside geometric and graph EDA.
2. Topology models — complete the frozen-geometry GCN (2a), followed by joint geometry + GCN training (2b); concurrently complete the shallow-topology models (1.5).
3. Preliminary results and extensions — produce initial analysis across Stages 1, 1.5, 2a and 2b; complete full UV-Net training and explore further architectures.
4. Complete analysis — compare all completed ladder stages through class-level and confusion analysis and learned-embedding visualization.
5. Delivery — complete final integration and reproducibility checks, report preparation and presentation recording.

References

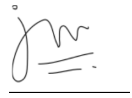
- [1] Jayaraman, P. K., et al. (2021). UV-Net: Learning From Boundary Representations. CVPR, 11703–11712.
[2] Cao, W., et al. (2020). Graph Representation of 3D CAD Models for Machining Feature Recognition With Deep Learning. ASME IDETC-CIE.

Team Contract

Task allocation: The ticks indicate intended areas of contribution rather than exclusive one-to-one ownership. Multiple members may work independently or collaboratively on the same item, after which implementations may be integrated or the most suitable selected. Items vary substantially in scope, effort and duration; workload should therefore be assessed across the project as a whole rather than by counting ticks. Responsibilities may be adjusted as implementation difficulty, dependencies and project priorities evolve.

Workstream	Task	Yuansen	Yong Jun	Kai Sheng	Ibrahim
Data preparation	Extraction & loading	✓		✓	
EDA	Geometry based		✓		✓
	Graph based		✓	✓	✓
Model construction and training	Train/test setup	✓		✓	✓
	Evaluation pipeline	✓		✓	✓
	Classifier	✓	✓	✓	✓
	CNN 2D face and 1D edge	✓	✓	✓	✓
	GCN & message passing	✓		✓	✓
	Shallow topology features		✓		
Result analysis & interpretation	Class level analysis		✓		
	Embedding visualization		✓	✓	
	Comparison across stages		✓		
Communication & integration	Code integration			✓	✓
	Weekly reporting		✓		
	Final report	✓			✓
	Slide outline and distribution		✓		

Signatures

Name	Signature
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